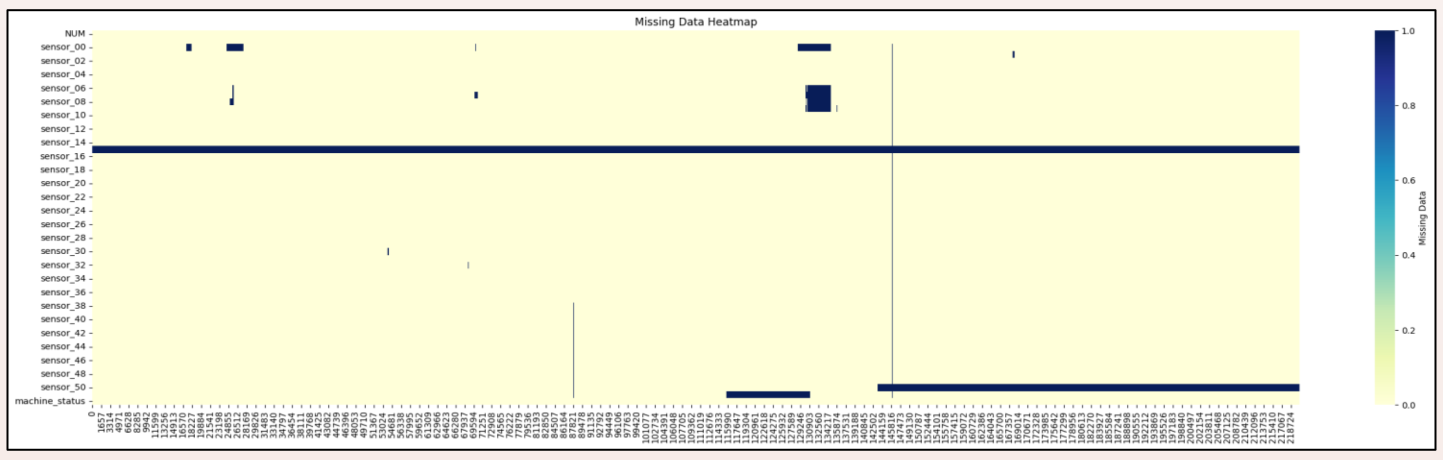
Anomaly Detection in Water Pump Sensor Data

By group 8
Assignment 2, TMM4128 Machine learning for engineers

Business Objective
Industrial facilities rely on the continuous operation of water pumps. Unexpected failures can lead to costly downtime and equipment damage.
Our goal is to build a model that detects abnormal behavior in sensor data — enabling **predictive maintenance** and reducing manual inspection needs.

1. Dataset
- Time-series data from industrial water pumps
- **52 numeric sensor features** (e.g., pressure, vibration, temperature)
- **Machine status labels**: NORMAL, RECOVERING, BROKEN
- ~220,000 total records

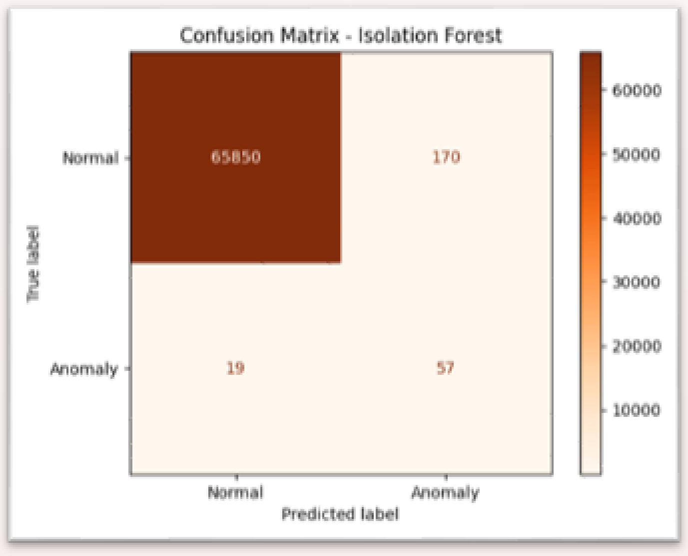
2. Data Preprocessing
- Dropped sensors with excessive missing values.
- Filled gaps using **forward-fill**, suitable for time series.
- Applied **StandardScaler** for normalization.
- Removed highly correlated sensors (correlation > 0.99).
- 70 / 30 train-test split.

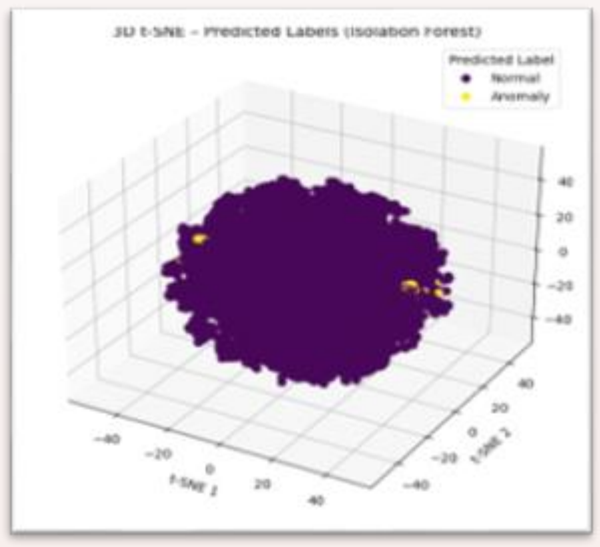


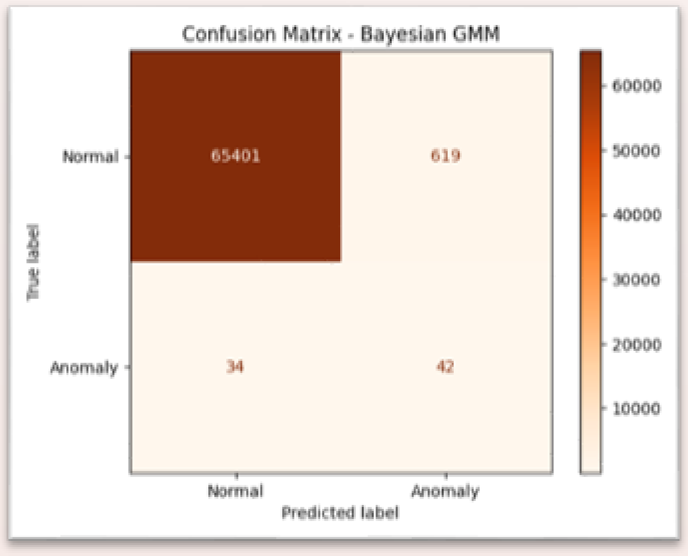
3. Problem Framing
- Treated as an **unsupervised anomaly detection** problem
- The model is trained on unlabeled data
- Ground truth labels (machine_status) are used only for evaluation
- We assume most data points are normal, with rare anomalies

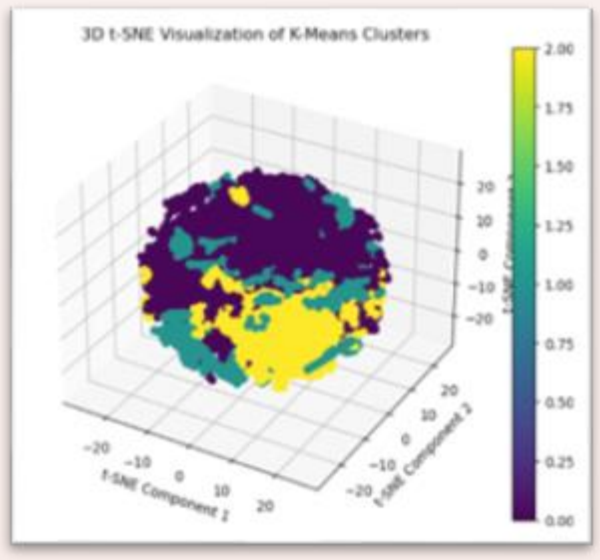
5. Tested and best Model
- **Isolation Forest:** *estimators*=80, *contamination*=0.12
- **Bayesian GMM:** *n_components*=3, *weight_concentration_prior* =0.1
- **K-Means:** *n_clusters* =3, init = "k-means++"

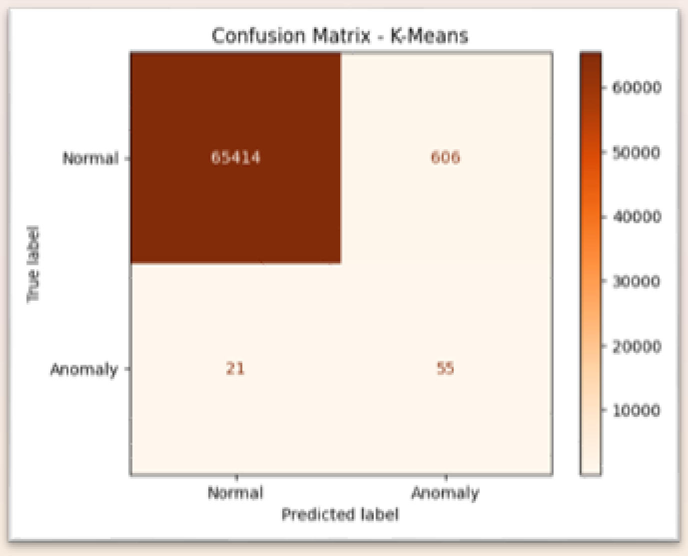
5. Evaluation
- Metrics used: **Precision, Recall, F1-score**
- Visualized results using **Confusion Matrix** and **t-SNE**

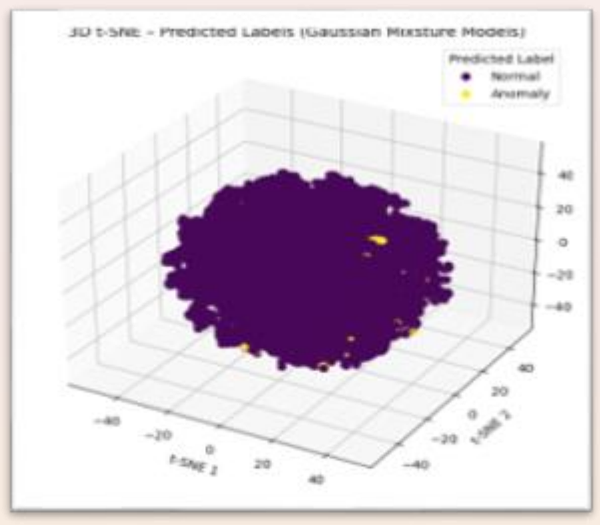












Model	Accuracy	Precision	Recall	F1-Score
Isolation Forest	1.00	0.25	0.75	0.38
Bayesian GMM	0.99	0.06	0.55	0.11
K-Means	0.99	0.08	0.72	0.15

Conclusion We built an unsupervised anomaly detection pipeline for sensor data, but all models struggled to detect rare failures.

The main issue was the **extremely imbalanced dataset** — very few BROKEN cases and many tightly clustered NORMAL samples. This made it hard to form meaningful clusters and caused models to show high accuracy while missing actual anomalies.

Key takeaways:

•Isolation Forest performed best but still missed most anomalies.

•PCA helped reduce noise.

•Accuracy was misleading — recall and precision for anomalies were low.

Recommendations:

•Try **semi-supervised** methods or anomaly-aware models like autoencoders.

•Address **class imbalance** and explore time-based features or domain knowledge.